Distributing Objects into Boxes

- How many ways are there to deal 5 cards to each of 4 players, from the standard deck of 52 cards?
- **Solution**
- Here, the cards are the distinguishable objects and the players hands are the distinguishable boxes.
- The first player's can be chosen in $C(52,5)$ ways
- The second player: $C(47,5)$ ways
- The third player: $C(42,5)$ ways
- The fourth player: $C(37,5)$ ways
- By the product rule, the total number of ways is

$$C(52,5)C(47,5)C(42,5)C(37,5) = \frac{52!}{47!5!} \cdot \frac{47!}{42!5!} \cdot \frac{42!}{37!5!} \cdot \frac{37!}{32!5!} = \frac{52!}{5!5!5!5!32!}$$